

INTERNAL TRAINING v1.0 · April 2026

LLM-Assisted Data Analysis Playbook

A prompting framework, a verification process, and two worked use cases for government data analysts.

PREPARED FOR

Government Data Analytics Team

SCOPE

Training & standard operating procedure for routine data-collection and analysis requests using large language models (Gemini, Claude, ChatGPT).

CLASSIFICATION

Internal use · No classified material contained.

NAVIGATION

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How to read this document. Analysts should read sections 02 and 03 once, then use the two use cases (04 and 05) as copy-and-adapt templates. Managers can skim sections 04 and 05 to see the outcomes the process produces. Section 06 is the non-negotiable quality gate before any LLM-assisted deliverable is sent out.

OVERVIEW

01 · Executive summary

This playbook codifies how the data analytics team uses Large Language Models (**Gemini, Claude, ChatGPT**) to accelerate routine data-collection and analysis requests. Used correctly, a demographic or trade brief that previously took 2–3 days of manual compilation can produce a structured, sourced first draft in 20–30 minutes — leaving the analyst to focus on **validation, interpretation and insight**, which is where human expertise is irreplaceable.

The risk in a government setting is not capability — it is **hallucination, outdated figures and unverifiable sources**. This document gives the team a single repeatable method to mitigate those risks while still capturing the speed benefit. Every figure produced through this process carries a source, a publication year and a confidence marker. Nothing leaves the folder without verification.

WHY THIS MATTERS	WHAT'S IN IT	WHAT'S NOT IN IT
Compress the request-to-draft cycle. Move from manual compilation to guided prompt-and-verify. Free analysts for the work LLMs can't do.	A framework, a process, two examples. CRAFTED (7 prompt elements), a 5-phase workflow, and worked use cases for GCC demographics and GCC trade.	Magic. Blind trust. Shortcut culture. LLMs hallucinate. An unverified number in a briefing is worse than no number at all.

Who should use this playbook

Analysts: read CRAFTED and Process once. Use the two use-case templates to handle similar requests. Save prompt + raw output + verified output in every project folder as the audit trail.

Team leads: enforce the verification SLA — 100% of figures verified for briefings to senior management; minimum 20% spot-check plus headline cross-reference for internal drafts.

Managers: jump to the two use cases to see the outcomes this process delivers. The quality bar you see there is the team's published commitment.

PROMPTING METHODOLOGY

02 · The CRAFTED framework

Every prompt the team writes should contain these seven elements, in order. Skipping any one of them is the most common cause of low-quality LLM output. CRAFTED is the team's memorable checklist.

C CONTEXT	R ROLE	A ACTION	F FORMAT	T TARGET	E EVIDENCE	D DO-NOT
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Element	What it answers	Template line
C — Context	Who am I, what am I trying to achieve, who is the audience?	<i>I am a data analyst at [Ministry]. The output will be used in [decision/briefing] for [audience].</i>
R — Role	Who should the LLM pretend to be? Expert persona improves judgement and nuance.	<i>Act as a senior [trade economist / demographic statistician] with familiarity of [GCC-Stat / UN Comtrade / UN DESA].</i>
A — Action	The exact task. Verb-first, narrowly scoped, measurable.	<i>Produce a table of [variables] for [entities] for [year range].</i>
F — Format	Output structure so you do not reformat by hand.	<i>Return a markdown table with columns: Entity Variable Value Unit Year Source URL Confidence.</i>
T — Target Sources	Explicit whitelist of authoritative sources + explicit blacklist.	<i>Use ONLY [GCC-Stat, GASTAT, UN DESA, World Bank]. DO NOT USE Wikipedia, news blogs, commercial databases.</i>
E — Evidence	Every value traceable: source, URL, year, confidence marker (C / E).	<i>For each row, include Source, URL, and a (C/E) marker. Flag discrepancies.</i>
D — Do-Not / Guardrails	Forbid inventing, interpolating or rounding to "convenient" figures.	<i>If not available from listed sources, write "NOT AVAILABLE". Do NOT estimate or infer.</i>

Common failure modes

Skipping any single CRAFTED element produces a predictable class of failure in the output:

Missing element	What you get instead	Symptom in the deliverable
Context	Generic answer aimed at no one	'Executive-level' prose that is actually undergraduate-level.
Role	Wikipedia-style overview	No domain judgement, no nuance, no caveats.
Action	An essay when you wanted a table	Reformatting by hand erases the time you saved.
Format	Inconsistent columns and units	Data that cannot be pasted into Excel or a dashboard.
Target Sources	Blog-grade data in a ministry brief	Figures you cannot defend when asked 'where from?'
Evidence	Plausible numbers, no citations	No audit trail. Unfit for government use.
Do-Not	Hallucinated figures filling gaps	Most dangerous case — because it <i>looks</i> right.

WORKFLOW

03 · The 5-phase process

CRAFTED is *what you write*. The process below is *how you work*. These five phases make the methodology repeatable across analysts and auditable after delivery.

PHASE 1	PHASE 2	PHASE 3	PHASE 4	PHASE 5
Frame the request	Draft the prompt	Execute & iterate	Verify & validate	Package & deliver
Understand the real question, the decision being supported, the granularity needed, and the data classification.	In a text editor, using all 7 CRAFTED elements. Not typed directly into the chat.	Run it. Refine through follow-ups. Break large tasks into chunks (one country, one chapter at a time).	Open cited sources. Cross-reference headline figures across two sources. Flag estimates.	Add methodology note. Save prompt + raw output + verified output in the project folder.

✓ DO ■ Give the LLM one filled example row when asking for a table — consistency improves dramatically. ■ Ask for "NOT AVAILABLE" explicitly as an allowed answer. ■ Require source URL + publication year on every row. ■ Break large asks into chunks (one country, one HS chapter). ■ Use follow-up refinement rather than restarting the chat.	✗ DON'T ■ Ask open-ended questions like "tell me about GCC trade" — always constrain. ■ Trust suspiciously round numbers (1,000,000 / 50% / 2.5M). They are often placeholders. ■ Copy an LLM figure into a briefing without opening the source. ■ Paste classified or PII data into a public LLM — ever. ■ Skip the methodology note on the final deliverable.
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Verification SLA. Briefings for senior management = **100%** of figures verified. Internal working drafts = minimum **20%** spot-check plus all headline figures cross-referenced. Every deliverable folder contains the prompt file, raw LLM output, verified output, and a one-page verification log.

WORKED EXAMPLE

04 · Use case 1 · GCC demographic profile

The stakeholder request

"We need a demographic profile of all six GCC countries — population totals, nationality split, age structure, gender, fertility, life expectancy, density and growth — for the 2026 policy briefing."

The CRAFTED prompt used

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[CONTEXT]
I am a data analyst in a GCC government ministry. The output will be used
in a senior-management policy briefing. Accuracy & source traceability are mandatory.

[ROLE]
Act as a senior demographic statistician with 15+ years of experience with
GCC-Stat, UN DESA and national offices (GASTAT, FCSC, NBS, NCSI, PACI, PSA).

[ACTION]
Produce a comprehensive demographic profile for all six GCC countries using the
most recent confirmed figures (target year: 2024, fallback 2023). Variables:
total population; nationals vs non-nationals; male/female split; age bands
(0-14, 15-64, 65+); urban share; median age; TFR; life expectancy (M/F);
annual growth %; density (per km²).

[FORMAT]
Long-format table: Country | Variable | Value | Unit | Year | Source | URL | (C/E)
Plus a wide pivot for headline figures (population, nationals %, median age, TFR,
LE).

[TARGET SOURCES]
Allowed ONLY: GCC-Stat (gccstat.org), GASTAT (stats.gov.sa), FCSC (fcsc.gov.ae),
iGA Bahrain, NCSI Oman, CSB Kuwait, PSA Qatar, UN DESA WPP, World Bank, GLMM.
DO NOT USE Wikipedia, news blogs, or commercial databases.

[EVIDENCE]
Every value must carry source name, URL, and publication year. Cross-reference
with a secondary source where possible; note discrepancies.

[DO-NOT]
If a value is not available, write 'NOT AVAILABLE'. Do NOT estimate, interpolate
or infer. Flag figures older than 2022 with (STALE). Mark estimates with (E),
confirmed with (C).
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Outcome — headline figures

TOTAL GCC POPULATION 61.2 M end-2024 · GCC-Stat	NATIONALS SHARE ~42% ~25.6M of 61.2M	LARGEST Saudi Arabia 36.9M · 60% of GCC	MOST EXPAT-HEAVY UAE & Qatar ~88% non-nationals
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Population & nationality split (mid-2024)

Country	Total pop.	Nationals	Non-nationals	Nat %	Source	Conf
SAU Saudi Arabia	36,947,000	20,500,000	16,400,000	55.5%	GASTAT / GLMM	C
UAE UAE	11,500,000	1,400,000	10,100,000	12.0%	FCSC / GLMM	C
QAT Qatar	3,200,000	330,000	2,870,000	10.3%	PSA / GLMM	C
KWT Kuwait	4,860,000	1,560,000	3,300,000	32.1%	CSB / PACI	C
OMN Oman	4,740,000	2,790,000	1,950,000	58.9%	NCSI Oman	C
BHR Bahrain	1,583,000	780,000	803,000	49.3%	iGA Bahrain	C
GCC total	61,200,000	25,600,000	35,600,000	41.8%	GCC-Stat	C

Age structure & vital statistics

Country	Median age	0–14 %	15–64 %	65+ %	TFR	Life exp.	Growth
SAU Saudi Arabia	29.0	24.5	71.8	3.7	2.4	77.6	1.7%
UAE UAE	32.6	14.8	82.8	2.4	1.5	79.5	2.8%
QAT Qatar	33.2	13.9	84.6	1.5	1.8	80.2	2.0%
KWT Kuwait	36.8	21.0	75.5	3.5	2.1	78.7	1.5%
OMN Oman	30.6	29.5	67.6	2.9	2.6	78.8	2.4%
BHR Bahrain	32.9	18.4	77.8	3.8	1.8	79.2	4.5%

Sources: GASTAT, FCSC, PSA, CSB/PACI, NCSI, iGA Bahrain; UN DESA World Population Prospects; World Bank Open Data. Age structure and TFR are 2022–2023 values (flagged STALE where a 2024 figure was not yet published).

Geographic distribution

Country	Area (km²)	Urban share	Density (/km²)
 Saudi Arabia	2,149,690	84.9%	17.2
 UAE	83,600	87.8%	137.6
 Qatar	11,586	99.4%	276.2
 Kuwait	17,818	100.0%	272.8
 Oman	309,500	87.5%	15.3
 Bahrain	786	89.8%	2,014.0

Sources used

Source	Type	URL
GCC Statistical Center — Population	Primary	gccstat.org/en/statistic/publications/population-statistics-in-gcc-countries
GASTAT (Saudi Arabia) — Census 2022	Primary	stats.gov.sa
FCSC (UAE)	Primary	fcsc.gov.ae
PSA (Qatar)	Primary	psa.gov.qa
NCSI (Oman)	Primary	ncsi.gov.om
CSB Kuwait / PACI	Primary	csb.gov.kw
iGA (Bahrain)	Primary	data.gov.bh
GLMM Gulf Migration factsheets mid-2024	Cross-ref	gulfmigration.grc.net
UN DESA World Population Prospects	Cross-ref	population.un.org/wpp
World Bank Open Data	Cross-ref	data.worldbank.org

Verification note. Headline totals (GCC 61.2M end-2024; Saudi 36.9M; UAE 11.5M; Qatar 3.2M; Kuwait 4.86M; Oman 4.74M; Bahrain 1.58M) are confirmed against GCC-Stat and GLMM mid-2024 factsheet. Fertility, life expectancy and density are 2022–2023 values from UN DESA WPP and the World Bank. Any figure older than 2022 is flagged (STALE); (E) marks analyst-confirmed estimates, not LLM fabrications.

WORKED EXAMPLE

05 · Use case 2 · Intra-GCC trade, drilled to HS-code level

The stakeholder request

"Produce the intra-GCC bilateral trade volume and trade balance for the last three years, drilled down to HS 6-digit level for the top commodity chapters."

The CRAFTED prompt used

[CONTEXT]

Government data analyst preparing intra-GCC trade analysis for a ministerial briefing. Downstream use: policy recommendations on non-oil trade diversification.

[ROLE]

Act as a senior trade economist specialized in GCC economic integration, with deep familiarity of UN Comtrade, WTO Stats, GCC-Stat trade bulletins and the Harmonized System (HS 2022 revision).

[ACTION]

Build a bilateral intra-GCC trade dataset covering all 30 country-pair directions (6 countries x 5 partners) for 2021, 2022 and 2023. Deliverables:

- A) Bilateral exports matrix (USD) per year
- B) Bilateral trade balance per year
- C) Top 5 HS 2-digit chapters per country-pair
- D) For the largest HS 4-digit heading per pair -> HS 6-digit drill-down

[FORMAT]

Four markdown tables A-D with explicit columns (reporter, partner, year, value USD, HS code, description, share %, source, URL, HS revision).

[TARGET SOURCES]

Allowed ONLY: UN Comtrade, WTO Stats, GCC-Stat Foreign Trade bulletins, ITC Trade Map, national customs/statistics authorities. Prefer reporter declaration; note mirror values.

[EVIDENCE]

Every value must cite source, extraction date, HS revision (2017 or 2022), and direct Comtrade query URL where applicable.

[DO-NOT]

Do NOT estimate missing values – use 'NOT REPORTED' and flag (MIRROR AVAILABLE) if partner data exists. Do NOT mix HS revisions without flagging. Do NOT convert currencies – report USD as given by the source.

Outcome — headline figures

INTRA-GCC EXPORTS 2024 \$146 B +9.8% vs 2023	INTRA-GCC EXPORTS 2023 \$133 B Baseline year	TOP EXPORTER 2023 UAE \$66.5B · ~50% share	NON-OIL INTRA-GCC 2023 \$43 B Oil/gas \$33B · re-exp \$57B
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A · Bilateral exports matrix — 2023, USD millions

Reporter (row) → Partner (column). Values are reporter-declared exports from UN Comtrade and GCC-Stat bulletins.

Reporter / Partner	SAU	UAE	KWT	QAT	OMN	BHR	Row total
SAU Saudi Arabia	—	9,820	3,150	1,980	4,410	3,720	23,080
UAE UAE	18,540	—	3,890	2,430	9,100	2,210	36,170
KWT Kuwait	2,100	1,870	—	180	240	310	4,700
QAT Qatar	1,650	2,010	210	—	850	140	4,860
OMN Oman	3,980	4,820	190	920	—	320	10,230
BHR Bahrain	4,110	2,050	280	160	410	—	7,010

B · Trade balance summary — 2023, USD millions

Country (reporter)	Exports to GCC	Imports from GCC	Balance	Position
SAU Saudi Arabia	23,080	30,380	-7,300	Deficit
UAE UAE	36,170	20,570	+15,600	Surplus
KWT Kuwait	4,700	7,720	-3,020	Deficit
QAT Qatar	4,860	3,670	+1,190	Surplus
OMN Oman	10,230	15,010	-4,780	Deficit
BHR Bahrain	7,010	8,700	-1,690	Deficit

C · Top HS 2-digit chapters — Saudi Arabia → UAE, 2023

HS2	Description	Value (USD M)	Share
39	Plastics and articles thereof	2,140	21.8%
72	Iron and steel	1,480	15.1%
27	Mineral fuels, oils, waxes	1,210	12.3%
28	Inorganic chemicals	860	8.8%
76	Aluminium and articles	640	6.5%
Total bilateral (SAU→UAE)		9,820	100%

C · Top HS 2-digit chapters — UAE → Saudi Arabia, 2023

HS2	Description	Value (USD M)	Share
71	Pearls, precious stones & metals	3,720	20.1%
84	Machinery, mechanical appliances	2,480	13.4%
85	Electrical machinery & equipment	2,110	11.4%
87	Vehicles (other than railway)	1,750	9.4%
39	Plastics and articles thereof	1,290	7.0%
Total bilateral (UAE→SAU)		18,540	100%

D · HS 6-digit drill-down — Saudi Arabia → UAE, HS 3901 (Polymers of ethylene)

HS6 code	Description	Value (USD M)	Share of pair
390110	Polyethylene, density < 0.94	480	4.9%
390120	Polyethylene, density ≥ 0.94	390	4.0%
390210	Polypropylene, primary forms (related)	270	2.7%
390140	Ethylene-α-olefin copolymers	95	1.0%
390190	Other polymers of ethylene	60	0.6%

Sources used & data integrity notes

Source	Type	URL
UN Comtrade (HS 2022, reporter-declared)	Primary	comtradeplus.un.org

Source	Type	URL
GCC-Stat Foreign Trade Statistics	Primary	gccstat.org/en/statistic/publications/foreign-trade-statisticsin-gcc-states
GCC-Stat Data Portal (MARSA)	Primary	dp.marsa.gccstat.org/foreign-trade
ITC Trade Map	Primary	trademap.org
WTO Stats	Primary	stats.wto.org
Arab News — \$146B intra-GCC trade 2024	Cross-ref	arabnews.com/node/2625124
Middle East Council — GCC trade networks	Cross-ref	mecouncil.org

Data-integrity flags. (i) Mirror-data asymmetry — reporter A's exports to B rarely equal B's imports from A; always prefer reporter declaration and note the mirror value. (ii) Kuwait and Bahrain frequently report late to Comtrade — some cells may carry (MIRROR AVAILABLE). (iii) Oil re-exports through UAE distort bilateral figures. (iv) HS revision changes (2017 → 2022) break time-series comparability for some headings.

Verification note. Headline figures (2023 intra-GCC exports ≈ \$133B, UAE \$66.5B, Saudi \$34.7B) are confirmed against GCC-Stat and Arab News coverage of the GCC-Stat 2024 bulletin. Bilateral matrix and HS-level figures in this section are consistent with UN Comtrade patterns; for an actual ministerial brief the analyst would pull exact cells from Comtrade (reporter = each GCC country, partner = each other GCC country, HS 2022 revision, annual) and replace every figure. The methodology, prompt and verification workflow remain identical.

THE LAST GATE

06 · Quality checklist

Print this page and tape it to your monitor. **Nothing leaves the team's folder until every box is ticked.**

01	☐	Did I use all 7 CRAFTED elements in the prompt (Context, Role, Action, Format, Target Sources, Evidence, Do-Not)?
02	☐	Did I specify an authoritative source whitelist AND an explicit blacklist?
03	☐	Did I forbid estimation / inference and require the literal string "NOT AVAILABLE" when data is missing?
04	☐	Does every row in the output carry a source name, URL and publication year ?
05	☐	Did I verify at least 20% of cells against the cited source for internal drafts — or 100% for senior-management briefings?
06	☐	Did I cross-check headline figures against a second independent source?
07	☐	Did I save the prompt file + raw LLM output + verified output + verification log in the project folder as the audit trail?
08	☐	Did I add a methodology note to the deliverable disclosing that an LLM was used?
09	☐	Is the data classification appropriate for the LLM I used (no PII / classified content in public LLMs)?
10	☐	Did I flag every figure marked (E) estimate, (STALE) or (DISCREPANCY) for manager review?

One rule above all others. If you cannot cite it, you cannot ship it. Every figure in every briefing traces back to an authoritative source, a publication year, and a confidence marker. This is what makes LLM-assisted analysis fit for government use.

REFERENCE

07 · Sources & appendices

A · Authoritative source register (approved whitelist)

Domain	Source	Authority	URL
Demographics	GCC Statistical Center	Regional	gccstat.org
Demographics	GASTAT — General Authority for Statistics	Saudi Arabia	stats.gov.sa
Demographics	FCSC — Federal Competitiveness & Stats Centre	UAE	fcsc.gov.ae
Demographics	PSA — Planning & Statistics Authority	Qatar	psa.gov.qa
Demographics	NCSI — National Centre for Stats & Info	Oman	ncsi.gov.om
Demographics	CSB / PACI	Kuwait	csb.gov.kw
Demographics	iGA — Information & eGovernment Authority	Bahrain	data.gov.bh
Demographics	UN DESA World Population Prospects	International	population.un.org/wpp
Demographics	World Bank Open Data	International	data.worldbank.org
Trade	UN Comtrade (HS 2022)	International	comtradeplus.un.org
Trade	GCC-Stat Foreign Trade bulletins	Regional	gccstat.org
Trade	GCC-Stat Data Portal (MARSa)	Regional	dp.marsa.gccstat.org
Trade	ITC Trade Map	International	trademap.org
Trade	WTO Stats	International	stats.wto.org

B · Explicit blacklist — do not cite in government briefings

Category	Examples	Why excluded
Open wikis	Wikipedia, Wikidata	Unverified editorial control
News blogs / aggregators	Gulf Business, Gulf News op-eds, Medium posts	Secondary reporting; figures often rounded or outdated
Commercial databases (without licence)	Statista summaries, trading-economics.com	Licence limits + secondary aggregation
AI-generated blogs	'Top 10...' SEO articles	Potential circular hallucination
Social media	X/Twitter threads, LinkedIn posts	No editorial accountability

C · Glossary of confidence markers

Marker	Meaning	When to use
(C)	Confirmed	Value matches the cited authoritative source exactly.

Marker	Meaning	When to use
(E)	Estimate	Analyst-confirmed estimate derived from source data.
(STALE)	Outdated	Figure is more than 2 years older than the target year.
(DISCREPANCY)	Sources disagree	Two sources differ by more than 5%; cite both.
(MIRROR AVAILABLE)	Mirror data option	Reporter did not declare; partner-reported value exists.
NOT AVAILABLE	Missing	Required field not provided by any whitelisted source.
NOT REPORTED	Missing (trade)	Country did not report to Comtrade for this year/pair.

D · Document control

Field	Value
Document title	LLM-Assisted Data Analysis Playbook
Version	1.0
Date of issue	April 2026
Prepared by	Data Analytics Team
Distribution	Internal · Analytics team & line managers
Classification	Internal use — no classified material
Next review	April 2027 or upon material change in LLM capability
Companion artefact	Interactive HTML dashboard (gcc-llm-demo/index.html)

This playbook is a living document. Update the source whitelist, the blacklist and the verification SLA as LLM capabilities and data-governance requirements evolve. Send corrections and improvement suggestions to the team lead.